# A Generalized Nash Equilibrium Model for Competing Plastic-Recycling Agents: Formulation, Parameterization, and Base-Case Results

**(Linear-Cost Interim Draft)**

Technical Report — Paola_Game_Theory project

October 10, 2026

**Color key used throughout this report:**
**[PAPER]** — value taken directly from Muñoz-Briones et al. or a table/figure therein.
**[DERIVED]** — value recovered from the authors' own underlying model (a Julia/JuMP notebook and a source spreadsheet feeding the paper) that is not printed in the paper itself, but is real data, not a guess.
**[ASSUMED]** — value with no traceable source in the paper or the authors' data; a defensible default chosen for this study and flagged for confirmation.

**What this draft is.** This is an *interim* report. Every agent's operating cost is modeled as **purely linear** in throughput (cost = $a_i q_i$, a constant cost per ton) — nothing convex or "calibrated" is involved. A convex refinement is a possible future extension, mentioned once, at the end (Section 5), and nowhere else in this document.

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# 1 Problem Statement

## 1.1 The source paper

Muñoz-Briones, Ikegwu, Martinez, Granger, Van Lehn, Zavala, Huber, and Avraamidou, *"Decision support for circular economy waste management of mixed plastic streams: A case study on post-industrial multilayer films"* (referred to throughout this report as "the paper"), solves a **single-planner, multi-objective mixed-integer nonlinear program (MINLP)** for the management of 8,000 tons/year of post-industrial multilayer plastic film generated in Wisconsin. A single decision-maker chooses, for the entire feedstock stream, which technology pathway to route it through — solvent-targeted recovery and precipitation (STRAP), incineration, gasification (three configurations), or landfill — and in the case of STRAP, which polymer to remove first and with which solvent. The planner's objective is one of three criteria evaluated separately: maximize profit, minimize greenhouse-gas emissions, or maximize a circularity index. The model is a single centralized optimization: there is one decision-maker and no competition for feedstock.

## 1.2 The extension: from a planner to a market

This project reformulates the same underlying technologies as a **Generalized Nash Equilibrium Problem (GNEP)**. Instead of one planner routing all 8,000 tons, each candidate technology is recast as an **independent, self-interested firm** that must *buy* feedstock from a common, limited pool and decide its own throughput to maximize its own profit. Three technologies are modeled as competing agents:

- **S — STRAP** (solvent-targeted recovery and precipitation), internal configuration frozen to the paper's best-performing pathway (Section 2.2).
- **W — Incineration** (waste-to-energy).
- **G — Gasification** for energy recovery.

Landfill is *not* an agent. It appears only in two places: (a) as an internal residue-disposal cost inside every agent's own objective function (each agent pays a tipping fee on the fraction of its own feed that it cannot convert to product), and (b) as the sink for whatever feedstock no agent chooses to buy, which affects system-wide emissions accounting but never any agent's profit.

Because all three agents draw from the *same* 8,000 t/yr pool, their individually optimal decisions are coupled through one shared constraint. This coupling through a shared feasible-region constraint — rather than through the agents' objective functions directly — is precisely what makes the game *generalized* rather than an ordinary (Nash) game, and it is what this report's entire mathematical apparatus is built to handle.

## 1.3 The central question

When three firms compete for one limited pool of feedstock, competition itself implicitly prices that feedstock: every additional ton any one firm buys is a ton the others cannot have. That implicit price is the **shadow price** $\lambda$ (USD/ton) on the shared feedstock constraint. $\lambda$ is never chosen or assumed — it is an emergent output of the equilibrium, and it is the headline quantity this entire study is built to compute. Economically, $\lambda$ acts as a uniform surcharge that every agent pays on top of its own production costs, driving each firm's optimal throughput down from what it would choose if feedstock were free and unlimited.

### 1.4 Scope of this report

This is an **interim draft** reporting two results, both under a **linear** operating-cost structure:

1. The base-case equilibrium at the paper's own feedstock level, $Q_{\text{tot}} = 8{,}000$ t/yr (Section 4.1).
2. **SA-1**: how the equilibrium price $\lambda$ responds to feedstock scarcity, sweeping $Q_{\text{tot}}$ from 0 to 18,000 t/yr (Section 4.2).

Further sensitivity work (whether the assumed capacities are themselves defensible, and how much the most load-bearing assumed parameters move the headline number) and a convex-cost refinement are possible future extensions, noted at the end, in Section 5.

**Explicitly out of scope** for this version of the model itself (not just this draft): output-price coupling (all prices are fixed), cost coupling (e.g. a landfill fee that depends on total system-wide residue volume), an emissions cap, and any genuine two-stage (endogenous-capacity) equilibrium. These are deliberate boundaries, not oversights, and are revisited in Section 5.

## 2 Mathematical Formulation

### 2.1 Sets and indices

| Symbol | Meaning |
|---|---|
| $i \in \{S, W, G\}$ | Agent index: STRAP, Incineration, Gasification |
| $k$ | Product index (agent-specific; see Table 2) |

### 2.2 STRAP's frozen internal configuration

STRAP's process design is a superstructure in the paper: which polymer to remove in which wash, with which solvent, is itself a decision. For this GNEP, STRAP's internal configuration is **fixed** to the paper's own best-performing (highest-profit) pathway, so that STRAP's problem reduces to a single continuous decision (how much feedstock to buy) with no internal binary variables:

- Wash 1: remove PE using tetrachloroethylene (PCE).
- Wash 2: remove EVOH using $\gamma$-butyrolactone.
- Overall yield: 97%.

This choice is essential to the model's tractability: with no binary variables anywhere in the game, every agent's optimization problem is smooth, which is what makes stacking Karush–Kuhn–Tucker (KKT) conditions across agents (Section 2.7) a valid solution method.

### 2.3 Decision variables

Per agent $i$, the only free decision is

$$q_i \geq 0 \quad [\text{ton/yr}] \qquad \text{— feedstock purchased and processed by agent } i.$$

Every other quantity — product outputs, residue, emissions — is **affine in** $q_i$ and is declared as an explicit variable with its own defining constraint (rather than pre-substituted away), so the model structure generalizes cleanly to later extensions:

$$y_{ik} = \eta_{ik}\, q_i \qquad \text{[product-}k\text{ units/yr]} \tag{1}$$

$$r_i = \rho_i\, q_i \qquad \text{[ton/yr, residue to landfill]} \tag{2}$$

$$E_i = e_i\, q_i \qquad \text{[t CO}_2\text{e/yr, reported only, never constrained]} \tag{3}$$

where $\eta_{ik}$ is the yield of product $k$ per ton of feedstock processed by agent $i$, $\rho_i$ is agent $i$'s residue fraction, and $e_i$ is agent $i$'s total emission factor (process emissions plus the emissions of landfilling its own residue).

### 2.4 Agent $i$'s optimization problem

Each agent solves a profit-maximization problem over its own throughput $q_i$, taking the (yet unknown) shared feedstock price $\lambda$ as given. Operating cost is **linear** in throughput: $a_i$ is a constant cost per ton (Section 3.7), so total operating cost is simply $a_i q_i$.

$$\max_{q_i \ge 0} \quad \pi_i = \underbrace{\sum_k P_k\, \eta_{ik}\, q_i}_{\text{revenue}} - \underbrace{a_i\, q_i}_{\text{operating cost}} - \underbrace{p_{\text{raw}}\, q_i}_{\text{feedstock cost}} - \underbrace{c_{\text{lf}}\, \rho_i\, q_i}_{\text{residue disposal}} - \underbrace{\text{ACC}_i}_{\text{annualized CAPEX (constant)}} \tag{4}$$

subject to

$$q_i \le \text{Cap}_i \qquad \text{(private capacity bound; multiplier } \mu_i\text{)} \tag{5}$$

$$\sum_{j \in \{S,W,G\}} q_j \le Q_{\text{tot}} \qquad \text{(\textbf{shared} feedstock pool; multiplier } \lambda\text{)} \tag{6}$$

where $P_k$ is the fixed market price of product $k$ (USD per product unit), $p_{\text{raw}}$ is the price paid per ton of feedstock, $c_{\text{lf}}$ is the landfill tipping fee paid on the agent's own residue, and $\text{ACC}_i$ is the annualized capital charge — a constant that does *not* affect the equilibrium throughput $q_i^*$, but does determine whether the agent is economically viable net of capital.

### 2.5 Net margin

Collect every constant per-ton term of Eq. (4) into one number:

$$m_i \equiv \sum_k P_k\, \eta_{ik} - a_i - p_{\text{raw}} - c_{\text{lf}}\, \rho_i \qquad \text{[USD/ton]} \tag{7}$$

$m_i$ is agent $i$'s profit per additional ton of feedstock, and — because cost is linear — it is a *constant*: the same number at every throughput level, not just near zero. $\pi_i$ then reduces to $\pi_i = m_i q_i - \text{ACC}_i$, an affine (straight-line) function of $q_i$. $m_i$ is the single number that ranks the agents' willingness to pay for feedstock.

### 2.6 The shared constraint and the Variational Equilibrium

Because constraint (6) is shared by all three agents, this is a *Generalized* Nash game: each agent's own feasible region depends on what the *others* choose. Such games generally admit infinitely many Generalized Nash Equilibria (GNE), each corresponding to a different way of splitting the shared resource, and each potentially supported by a *different* multiplier on the shared constraint for each agent.

This study solves for the **Variational Equilibrium (VE)**: the refinement of GNE in which *every agent faces the same multiplier* $\lambda$ on the shared constraint. Economically, this is the natural solution concept for a competitive market: $\lambda$ is a single, uniform market-clearing price that every buyer faces identically, exactly as a real commodity market would produce.

### 2.7 Stacked KKT / complementarity system

Because $\pi_i$ is affine (hence concave, though not *strictly* concave) in $q_i$, its Karush–Kuhn–Tucker conditions are still necessary and sufficient for its own optimum given $\lambda$. Stacking all three agents' KKT conditions together with the complementarity condition on the shared constraint gives one square Mixed Complementarity Problem (MCP):

$$0 \;\le\; q_i \;\perp\; \lambda + \mu_i - m_i \;\ge\; 0 \qquad \forall i \in \{S, W, G\} \tag{8}$$

$$0 \;\le\; \mu_i \;\perp\; \mathrm{Cap}_i - q_i \;\ge\; 0 \qquad \forall i \in \{S, W, G\} \tag{9}$$

$$0 \;\le\; \lambda \;\perp\; Q_{\mathrm{tot}} - \sum_j q_j \;\ge\; 0 \tag{10}$$

where $a \perp b$ denotes the complementarity condition $a \cdot b = 0$. Note condition (8): the stationarity term $\lambda + \mu_i - m_i$ does *not* depend on $q_i$ — a direct consequence of linear cost. This is what makes the best response a step function rather than a smooth curve, worked out next.

### 2.8 Best response, and the tie case

Because $\lambda + \mu_i - m_i$ is constant in $q_i$, an unconstrained agent's profit $m_i q_i$ is monotonic in $q_i$ with no interior optimum: the agent wants either *all* the feedstock it can get, or *none*. Concretely, at a candidate price $\lambda$:

$$q_i(\lambda) \;=\; \begin{cases} \mathrm{Cap}_i & m_i > \lambda \quad \text{(wants strictly more; capacity binds)} \\ 0 & m_i < \lambda \quad \text{(priced out)} \\ \text{any value in } [0, \mathrm{Cap}_i] & m_i = \lambda \quad \text{(exactly indifferent)} \end{cases} \tag{11}$$

The third case is the only subtlety linear cost introduces: at the one price where an agent's margin exactly equals $\lambda$, *every* throughput is equally profitable for that agent, so its own optimization alone does not pin down a quantity. What *does* pin it down is the shared constraint: whichever agent is on the margin is handed exactly the residual feedstock needed to make $\sum_i q_i = Q_{\mathrm{tot}}$ — the standard economic convention for a marginal price-setting supplier under constant marginal cost. See Section 4.1 for exactly this happening in the base case.

### 2.9 Equilibrium states

At any solved equilibrium, each agent occupies exactly one of three mutually exclusive states, determined by *where it actually sits*, never by a static property of its own cost alone:

| State | Condition | Economic meaning |
|---|---|---|
| At Capacity | $q_i = \mathrm{Cap}_i,\ \mu_i > 0$ | Wants more feedstock, cannot have it |
| Price-Setter | $0 < q_i < \mathrm{Cap}_i,\ \mu_i = 0$ | On the margin; $m_i = \lambda$ pins the price |
| Priced Out | $q_i = 0,\ m_i < \lambda$ | Cannot outbid the others for a single ton |

At any given $Q_{\mathrm{tot}}$, at most one agent is normally the Price-Setter — it is that agent's own tie condition $m_i = \lambda$ that determines the market-clearing price for *everyone*, since a common $\lambda$ is exactly what makes the equilibrium variational.

### 2.10 Emissions accounting

Total system emissions include every agent's own process-plus-residue emissions *and* the emissions of landfilling whatever feedstock no agent buys:

$$E_{\mathrm{system}} \;=\; \sum_i e_i\, q_i^{*} \;+\; g_{\mathrm{lf}}\left(Q_{\mathrm{tot}} - \textstyle\sum_i q_i^{*}\right) \tag{12}$$

### 2.11 Solving for the equilibrium

Because every agent's best response is the step function of Eq. (11), the equilibrium does not require an iterative numerical solver here: rank the three agents by margin $m_i$ (descending), and hand each one its full capacity in that order until $Q_{\mathrm{tot}}$ is exhausted. The last agent whose capacity is only *partially* used is the Price-Setter, and $\lambda$ equals its margin $m_i$. Section 4.1 carries this out explicitly for the base case.

## 3 Parameters

### 3.1 Feedstock composition and base-case availability

| Symbol | Description | Value | Source |
|---|---|---|---|
| $Q_{\mathrm{tot}}^{\mathrm{base}}$ | Base-case feedstock availability | 8,000 ton/yr | **[PAPER]** |
| $f_{\mathrm{PE}}$ | Feedstock mass fraction, PE | 0.60 | **[PAPER]** |
| $f_{\mathrm{PET}}$ | Feedstock mass fraction, PET | 0.20 | **[PAPER]** |
| $f_{\mathrm{N6}}$ | Feedstock mass fraction, N6 (nylon) | 0.10 | **[PAPER]** |
| $f_{\mathrm{EVOH}}$ | Feedstock mass fraction, EVOH | 0.10 | **[PAPER]** |

PET and N6 are *not* recoverable by STRAP under the frozen configuration (Section 2.2): the paper states no solvent pair exists to separate a PET/N6 film, so that fraction of any STRAP-processed ton leaves as residue.

### 3.2 STRAP (agent S)

| Symbol | Description | Value | Source |
|---|---|---|---|
| $\eta_{\text{yield}}$ | Overall STRAP yield | 0.97 | [PAPER] |
| $P_{\text{PE}}$ | Recycled PE resin price | 1,173 USD/ton | [PAPER] |
| $P_{\text{EVOH}}$ | Recycled EVOH resin price | 8,100 USD/ton | [PAPER] |
| $\eta_{S,\text{PE}}$ | PE yield per ton feed ($= \eta_{\text{yield}} f_{\text{PE}}$) | 0.582 ton/ton | [DERIVED] |
| $\eta_{S,\text{EVOH}}$ | EVOH yield per ton feed ($= \eta_{\text{yield}} f_{\text{EVOH}}$) | 0.097 ton/ton | [DERIVED] |
| $\text{REV}_S$ | Gross revenue per ton feed | 1,468.386 USD/ton | [DERIVED] |
| $\rho_S$ | Residue fraction ($= 1 - \eta_{\text{yield}}(f_{\text{PE}} + f_{\text{EVOH}})$) | 0.321 ton/ton | [DERIVED] |
| $\text{CAPEX}_S$ | Total installed capital (wash 1 + wash 2 + shredder) | 53,645,274 USD | [DERIVED] |
| $\text{OPEX}_S^{\text{ref}}$ | Total operating cost at 8,000 t/yr reference basis | 2,490,259 USD/yr | [DERIVED] |
| $e_{\text{proc},S}$ | Process emission factor | 1.194 $\text{tCO}_2\text{e}$/ton | [DERIVED] |
| $\text{Cap}_S$ | Nameplate capacity | 6,000 ton/yr | [ASSUMED] |

**Provenance detail: STRAP's CAPEX and OPEX**

The paper itself prints no installed-cost or operating-cost figures for STRAP. These were recovered from the authors' own underlying BioSTEAM process simulation data (a Julia/JuMP notebook and its source spreadsheet, which feed the paper but are not printed in it), for the paper's best pathway (PE via PCE in wash 1, EVOH via $\gamma$-butyrolactone in wash 2), Energy Supply Scenario 1:

$$\text{CAPEX}_S = \underbrace{35{,}957{,}532}_{\text{wash 1, PE/PCE}} + \underbrace{14{,}557{,}669}_{\text{wash 2, EVOH/GBL}} + \underbrace{3{,}130{,}073}_{\text{shredder}} = 53{,}645{,}274 \text{ USD}$$

$$\text{OPEX}_S^{\text{ref}} = \underbrace{941{,}557}_{\text{wash 1}} + \underbrace{479{,}709}_{\text{wash 2}} + \underbrace{1{,}068{,}992}_{\text{shredder}} = 2{,}490{,}259 \text{ USD/yr, at 8,000 t/yr}$$

The process emission factor is likewise built from the per-resin BioSTEAM GWP factors (1.1326 $\text{tCO}_2\text{e}$/ton PE resin, 2.1661 $\text{tCO}_2\text{e}$/ton EVOH resin, 0.5575 $\text{tCO}_2\text{e}$/ton resin for the shredder), scaled to the 4,656 ton/yr of PE and 776 ton/yr of EVOH actually recovered at the 8,000 t/yr basis, and divided back onto a per-ton-of-feed basis: $e_{\text{proc},S} = 1.194$ $\text{tCO}_2\text{e}$/ton feed.

### 3.3 Incineration (agent W)

| Symbol | Description | Value | Source |
|---|---|---|---|
| $\text{REV}_W$ | Revenue (energy product value) per ton feed | 259.57 USD/ton | [DERIVED] |
| $\rho_W$ | Residue (ash) fraction | 0.15 ton/ton | [ASSUMED] |
| $e_{\text{proc},W}$ | Process emission factor | 2.38 $\text{tCO}_2\text{e}$/ton | [DERIVED] |
| $\text{Cap}_W$ | Nameplate capacity | 4,000 ton/yr | [ASSUMED] |
| $\text{OPEX}_W^{\text{ref}}$ | Operating cost per ton | 75 USD/ton | [ASSUMED] |
| $\text{CAPEX}_W$ | Installed capital at $\text{Cap}_W$ | 30,000,000 USD | [ASSUMED] |

**Provenance detail: incineration**

The paper never models incineration as a facility anyone owns and operates — it is treated purely as a *third-party off-take market*: the planner sells waste to incineration for 110 USD/ton (`Cgas_pw` in the authors' notebook), and the authors' spreadsheet records incineration OPEX and CAPEX as literally zero. Because this GNEP requires incineration to be an operating firm, its **revenue** was taken as $\text{REV}_W = 259.57$ USD/ton (`Cwte`, the authors' own figure for "cost of the product" — i.e. the value of the electricity/heat sold — listed alongside the STRAP resin prices in the same table), while its **operating cost** has *no* basis in the paper or the authors' data at all and was assumed at 75 USD/ton, a mid-range figure for US municipal-solid-waste waste-to-energy O&M (typical range 60–90 USD/ton). This is flagged as Question 1 and Question 2 for Paola (Section 6), since incineration turns out to be the agent that *sets* the equilibrium price at the base case (Section 4.1).

## 3.4 Gasification for energy recovery (agent G)

| Symbol | Description | Value | Source |
|---|---|---|---|
| $\text{REV}_G$ | Revenue (heat + electricity) per ton feed | 147.98 USD/ton | **[DERIVED]** |
| $\rho_G$ | Residue (ash/char) fraction | 0.08 ton/ton | **[ASSUMED]** |
| $e_{\text{proc},G}$ | Process emission factor | 1.07 $\text{tCO}_2\text{e}$/ton | **[DERIVED]** |
| $\text{Cap}_G$ | Nameplate capacity | 4,000 ton/yr | **[ASSUMED]** |
| $\text{OPEX}_G^{\text{ref}}$ | Operating cost per ton | 159.87 USD/ton | **[DERIVED]** |

**Provenance detail: gasification revenue**

$\text{REV}_G$ is built directly from the authors' own product yields and prices for gasification-for-energy-recovery:

$$\text{REV}_G = \underbrace{583.33\ \tfrac{\text{MJ}}{\text{ton}} \times 0.13\ \tfrac{\text{USD}}{\text{MJ}}}_{\text{heat}} + \underbrace{724.3693\ \tfrac{\text{kWh}}{\text{ton}} \times 0.0996\ \tfrac{\text{USD}}{\text{kWh}}}_{\text{electricity}} = 147.98\ \text{USD/ton}$$

Two of the underlying inputs are dimensionally suspect — the heat price of 0.13 USD/MJ (= 130 USD/GJ) is roughly an order of magnitude above typical industrial heat prices (5–15 USD/GJ), and the heat yield of 583 MJ/tonne is far below the heating value of plastic waste (30–40 GJ/tonne) — but both are used unmodified since they are the authors' own figures. The two suspected errors push revenue in opposite directions (a lower heat price reduces it, a higher heat yield raises it), so the net effect of correcting them cannot be signed in advance. This is Question 10.

## 3.5 Gasification CAPEX scaling

The authors' own notebook hard-codes gasification-for-energy-recovery annualized CAPEX at four discrete capacities. Fitting a six-tenths power law $\text{CAPEX}_G(\text{Cap}) = A\,\text{Cap}^n$ to two of these points gives $n = 0.6000$, $A = 15{,}749.6$, and this fit reproduces the other two authors' data points to within 0.1%:

| Capacity (t/yr) | Authors' value (USD/yr) | Fitted value (USD/yr) | Error |
|---|---|---|---|
| 2,400 | 1,680,303 | 1,679,096 | 0.07% |
| 3,200 | 1,996,874 | (fit point) | — |
| 7,200 | 3,248,331 | (fit point) | — |
| 8,000 | 3,460,308 | 3,459,341 | 0.03% |

Note that this column is already headed "CAPEX [USD/yr]" in the authors' own spreadsheet — it is *already annualized*, so the capital recovery factor CRF is **not** applied a second time for gasification, unlike for STRAP and incineration.

### 3.6 Shared market and cost parameters

| Symbol | Description | Value | Source |
|---|---|---|---|
| $p_{\text{raw}}$ | Feedstock purchase price | 0 USD/ton | **[ASSUMED]** |
| $c_{\text{lf}}$ | Landfill tipping fee | 60 USD/ton | **[ASSUMED]** |
| CRF | Capital recovery factor (10%, 20 yr) | 0.1175 /yr | **[ASSUMED]** |
| $g_{\text{lf}}$ | Landfill emission factor (0.0816 plus a 9.2-mile haul) | 0.0865 $tCO_2e$/ton | **[DERIVED]** |

$p_{\text{raw}}$. The paper hands its single planner 8,000 t/yr of feedstock at zero cost, so there is nothing to extract; $p_{\text{raw}} = 0$ is chosen for consistency with the paper. The authors' own model implicitly values this waste at 110 USD/ton in a different context (the incineration transfer price, `Cgas_pw`); post-industrial waste can plausibly carry a *negative* price in reality (the generator pays to have it hauled away).

$c_{\text{lf}}$. A representative Wisconsin municipal solid-waste landfill gate rate. Held *fixed*, i.e. not a function of total system-wide residue volume. The authors' own spreadsheet implies a landfill cost of only 7.836 USD/ton, roughly 8× below real Wisconsin gate rates, and is retained in the code as a cross-check constant.

CRF. A conventional 10% discount rate, 20-year asset life, computed as $\text{CRF} = i(1+i)^n/[(1+i)^n - 1]$. The paper's own convention is undiscounted straight-line annualization over 20 years ($\text{CRF}^{\text{paper}} = 0.05$), retained in the code as a cross-check constant. This choice affects only which agents are *viable* net of capital; it never affects the equilibrium throughput $q_i^*$ or $\lambda$ (Eq. (4) shows $\text{ACC}_i$ is a pure constant in the objective).

### 3.7 Operating cost per agent

$a_i$ (Eq. (4)) is each agent's operating cost per ton of feedstock processed. It is read off directly, with no modeling step in between:

- **Incineration and gasification:** $\text{OPEX}_i^{\text{ref}}$ (Sections 3.3–3.4) is already reported per ton, so $a_W = 75.00$ and $a_G = 159.87$ USD/ton, used as-is.
- **STRAP:** $\text{OPEX}_S^{\text{ref}} = 2{,}490{,}259$ USD/yr is a total cost at a reference throughput of $q_S^{\text{ref}} = 8{,}000$ t/yr (Section 3.2), so $a_S = \text{OPEX}_S^{\text{ref}}/q_S^{\text{ref}} = 311.28$ USD/ton — a unit conversion (total cost ÷ throughput), not an assumption or a fitted parameter.

Table 1: Operating cost and net margin per agent, base-case capacities.

| $i$ | $\mathrm{Cap}_i$ [t/yr] | $\mathrm{REV}_i$ [USD/t] | $a_i$ [USD/t] | $m_i$ [USD/t] | $\rho_i$ [t/t] | $e_i^{\mathrm{tot}}$ [$\mathrm{tCO_2e}$/t] | $\mathrm{ACC}_i$ [USD/yr] |
|---|---|---|---|---|---|---|---|
| S | 6,000 | 1,468.39 | 311.28 | 1,137.84 | 0.321 | 1.221 | 6,303,320 |
| W | 4,000 | 259.57 | 75.00 | 175.57 | 0.150 | 2.393 | 3,525,000 |
| G | 4,000 | 147.98 | 159.87 | −16.69 | 0.080 | 1.077 | 2,282,998 |

No other number enters $a_i$. Table 1 collects the result.

Note the ranking $m_S \gg m_W > 0 > m_G$: gasification's margin is **negative** under this linear cost. It costs more, on average, to run gasification than the energy it produces is worth, so gasification would not want to operate at *any* feedstock price $\lambda \geq 0$ — this is a direct, unavoidable reading of the raw numbers in Section 3.4, not an artifact of anything in this document.

### 3.8 Products

Table 2: Products sold by each agent, with prices $P_k$ and yields $\eta_{ik}$.

| Agent | Product $k$ | $P_k$ | $\eta_{ik}$ | Unit |
|---|---|---|---|---|
| S | Recycled PE resin | 1,173 USD/ton | 0.582 | ton product / ton feed |
| S | Recycled EVOH resin | 8,100 USD/ton | 0.097 | ton product / ton feed |
| W | Electricity + heat | 259.57 USD/ton feed | 1.000 | — (lumped) |
| G | Exported heat | 0.13 USD/MJ | 583.33 | MJ / ton feed |
| G | Exported electricity | 0.0996 USD/kWh | 724.37 | kWh / ton feed |

### 3.9 Emissions unit note

The paper's Table 8 reports the landfill-only pathway's emissions as "652.80 kg $CO_2$-eq/year" against 8,000 t/yr of feedstock — i.e. $\approx 1.9$ kg $CO_2$ per tonne of plastic, roughly 1,000× below standard plastic-waste LCA figures (incineration alone is typically 2–3 *tonnes* $CO_2$ per tonne of plastic). Tracing this to its source shows it is a **table label error, not a data error**: the underlying factor is $g_{\mathrm{lf}} = 0.0816$ $\mathrm{tCO_2e}$/ton, and $0.0816 \times 8{,}000 = 652.80$ reproduces the paper's number exactly — the data is genuinely in *tonnes*, and the paper's column header should read "t $CO_2$-eq/yr," not "kg." This is corroborated by the paper's own Table 6, which gives an emissions upper bound of 21,303.36 *tonCO2eq*, the same order of magnitude as the Table 8 entries. No 1,000× correction is applied anywhere in this model; the unit-scale parameter EMIS_UNIT_SCALE is retained at 1.0.

## 4 Results

### 4.1 Base-case equilibrium

At $Q_{\mathrm{tot}} = 8{,}000$ t/yr, rank the agents by margin (Table 1): $m_S = 1{,}137.84 \gg m_W = 175.57 \gg m_G = -16.69$. Fill capacity in that order:

- STRAP takes its full capacity, 6,000 t/yr (it would take more if it could).
- That leaves $8{,}000 - 6{,}000 = 2{,}000$ t/yr, well inside incineration's 4,000 t/yr capacity — so incineration is exactly on the margin: it is handed the residual, and its own tie condition sets the price:

$$\boxed{\lambda^* = m_W = 175.57 \text{ USD/ton}} \tag{13}$$

- Gasification's margin is negative even before $\lambda$ is added, so it takes 0 regardless of price.

Table 3: Base-case equilibrium, $Q_{\text{tot}} = 8{,}000$ t/yr.

| $i$ | $q_i^*$ [t/yr] | share | state | $\mu_i$ [USD/t] | op. profit [USD/yr] | net profit [USD/yr] | emissions [$tCO_2e$/yr] |
|---|---|---|---|---|---|---|---|
| S | 6,000.0 | 75.0% | At Capacity | 962.27 | 6,827,062 | 523,742 | 7,328.8 |
| W | 2,000.0 | 25.0% | Price-Setter | 0.00 | 351,140 | −3,173,860 | 4,785.9 |
| G | 0.0 | 0.0% | Priced Out | 0.00 | 0 | −2,282,998 | 0.0 |
| Σ | 8,000.0 | 100.0% | — | — | 7,178,202 | −4,933,116 | 12,114.8 |

Operating profit is simply $m_i q_i^*$ (e.g. $W$: $175.57 \times 2{,}000 = 351{,}140$). $\mu_S = m_S - \lambda = 1{,}137.84 - 175.57 = 962.27$ is STRAP's shadow price on its own capacity — how much its profit would rise per additional ton of nameplate capacity, holding $\lambda$ fixed.

**Unbought feedstock:** 0.0 t/yr (the shared constraint binds exactly: $\sum_i q_i^* = Q_{\text{tot}}$). **System emissions** including landfilled residue: 12,114.8 $tCO_2e$/yr.

### Reading the equilibrium

Only STRAP is viable after annualized capital charges; incineration and gasification both show negative profit net of $ACC_i$ even while incineration operates profitably on a cash basis. Gasification is priced out for a stronger reason than the other two agents' capacity limits: its margin $m_G = -16.69$ is negative on its own, so it would not want to operate even if it faced no competition for feedstock at all.

Capital is treated as already paid, because capacities are fixed inputs; that is why incineration operates at a cash profit even though it loses money after its assumed capital charge. The headline price therefore rests on the incinerator existing: with no incinerator, STRAP would still take 6,000 t/yr, 2,000 t/yr would go unbought, and $\lambda$ at 8,000 t/yr would be 0 rather than 175.57. Separately, STRAP's capital (53.6 million USD) comes from the 8,000 t/yr simulation but is charged against a 6,000 t/yr capacity; scaling it with the six-tenths rule would lower STRAP's capital charge from 6.30 to 5.30 million USD/yr and raise its profit after capital from 0.52 to about 1.5 million USD/yr. Neither point affects $\lambda$.

### A limitation worth flagging

The tie case in Eq. (11) says incineration is indifferent among *any* throughput from 0 to 4,000 once $\lambda = m_W$; 2,000 is picked only because it is the unique amount that clears the shared pool given STRAP's 6,000. That is fine here, because exactly one agent is ever on the margin. It would *not* be fine if two agents ever tied at the same margin simultaneously — linear cost gives no way to split the residual between them, since both would be indifferent to any split. That situation does not

arise in this base case, but it means this linear model is not guaranteed to give a unique *allocation* (only a unique $\lambda$ and total) at every possible $Q_{\text{tot}}$ — a limitation worth keeping in mind before this model is used for a broader sweep.

**Independent cross-check**

Routing all 8,000 t/yr to STRAP alone gives operating profit 9,102,749 USD/yr, against the paper's own Table 8 maximum-profit figure of 7,551,142 USD/yr — a ratio of 1.205. This is a plausibility check only, not a validation: the figure here is operating profit before capital charges, whereas the paper's is net of capital and includes a gasification pathway, so the two are not like-for-like.

## 4.2 SA-1: $\lambda$ versus feedstock scarcity

The base case above fixes $Q_{\text{tot}}$ at the paper's own 8,000 t/yr. SA-1 asks what happens to the competitive price of feedstock as that pool grows or shrinks: $Q_{\text{tot}}$ is swept from 0 to 18,000 t/yr and the equilibrium re-solved at every point.

**The result: $\lambda$ is a staircase, not a curve**

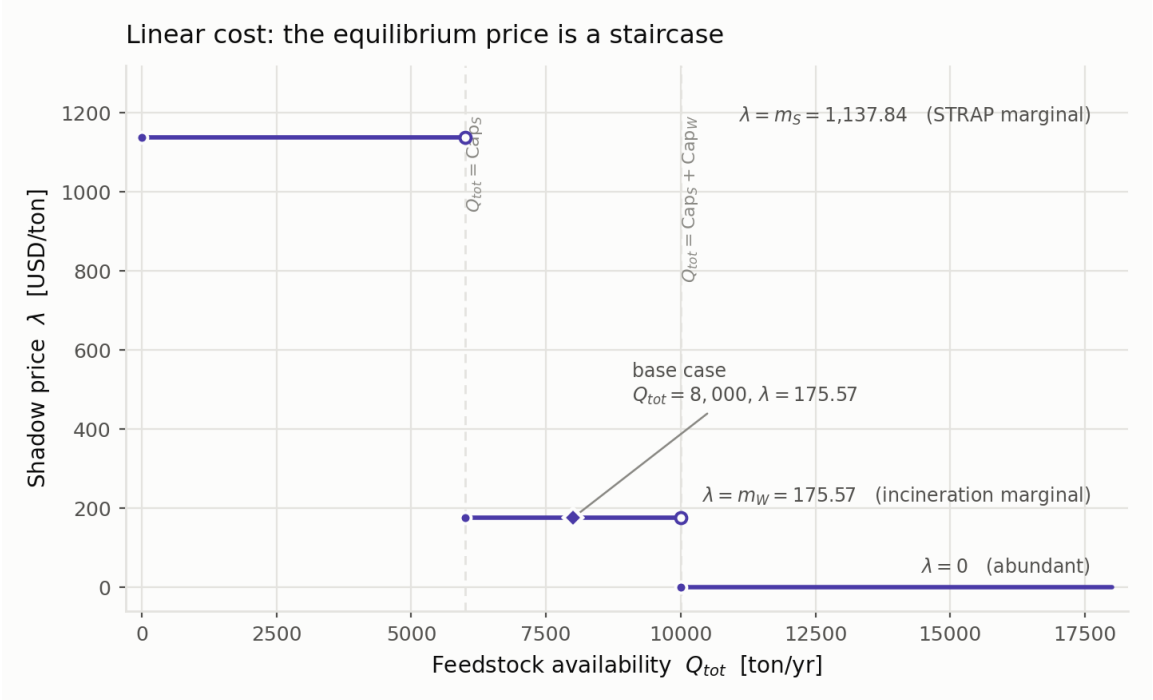



Figure 1: Equilibrium shadow price $\lambda$ against feedstock availability. Under linear cost the price is piecewise *constant*: each tread is attained at its left end (filled dot) and not at its right (open dot).

$\lambda$ takes exactly **three** values across the entire sweep (Table 4). This is a direct consequence of linear cost: whichever agent is on the margin has a *constant* marginal cost, so it posts one price and holds it for as long as it remains the marginal agent. The price changes only when the marginal agent itself changes.

Table 4: The three regimes of the linear-cost model. Each threshold is exact, not numerical.

| Regime | $Q_{\text{tot}}$ range [t/yr] | $\lambda$ [USD/t] | Who is on the margin |
|---|---|---|---|
| Severe scarcity | $0 \le Q_{\text{tot}} < 6{,}000$ | 1,137.84 | STRAP ($\lambda = m_S$) |
| Moderate scarcity | $6{,}000 \le Q_{\text{tot}} < 10{,}000$ | 175.57 | Incineration ($\lambda = m_W$) |
| Abundance | $Q_{\text{tot}} \ge 10{,}000$ | 0 | Nobody; feedstock is landfilled |

Both thresholds are capacity sums, exactly:

$$Q_{\text{tot}} = \text{Cap}_S = 6{,}000 \text{ t/yr} \qquad \text{(STRAP saturates; price drops 1,137.84} \to 175.57\text{)}$$
$$Q_{\text{tot}} = \text{Cap}_S + \text{Cap}_W = 10{,}000 \text{ t/yr} \qquad \text{(incineration saturates; price drops 175.57} \to 0\text{)}$$

The **abundance threshold is exactly** 10,000 **t/yr**, not 14,000 (total installed capacity), because gasification's margin is negative (Section 3.7): its 4,000 t/yr of capacity is never used at any availability, so it contributes nothing to aggregate demand.

### The price is not unique exactly at the two thresholds

At $Q_{\text{tot}} = 6{,}000$ every price in [175.57, 1,137.84] satisfies all equilibrium conditions, and at $Q_{\text{tot}} = 10{,}000$ every price in [0, 175.57] does. Throughout this report, and in the code, the *lowest* consistent price is reported at those two points, which is why each tread of Figure 1 is closed at its left end and open at its right.

### The first jump is the striking one

Just above $Q_{\text{tot}} = \text{Cap}_S$ the price falls by $m_S - m_W = 962.27$ USD/ton — an 85% collapse — in response to a vanishingly small amount of extra feedstock. The economics are stark: below 6,000 t/yr, STRAP can absorb everything available, so it is the marginal buyer and competition prices feedstock at *STRAP's* willingness to pay. The moment supply exceeds what STRAP can physically process, the marginal ton must go to the next buyer in line, and incineration's willingness to pay is roughly one-sixth of STRAP's. The price of feedstock is set entirely by whoever is last in the queue, and that identity changes discontinuously.

### Allocation: strict merit order

Agents fill in strict order of net margin, each running flat out before the next one takes anything — there is no range over which two agents share the marginal ton. This is the same bang-bang structure as Eq. (11), seen from the quantity side.

### What this does, and does not, tell us

**What it establishes.** The merit order and both switching thresholds are exact and robust: they depend only on the ranking of the margins $m_i$ and on the capacities, not on any curvature or shape assumption. The finding that gasification never participates at *any* feedstock level, and that the market saturates at 10,000 rather than 14,000 t/yr, are real results.

**What it cannot tell us.** Within a regime, $\lambda$ carries *no* information about how scarce feedstock actually is. At $Q_{\text{tot}} = 6{,}100$ t/yr (incineration barely started) and at 9,900 t/yr (incineration nearly saturated) the model returns the identical price, 175.57 USD/ton. A linear-cost model cannot

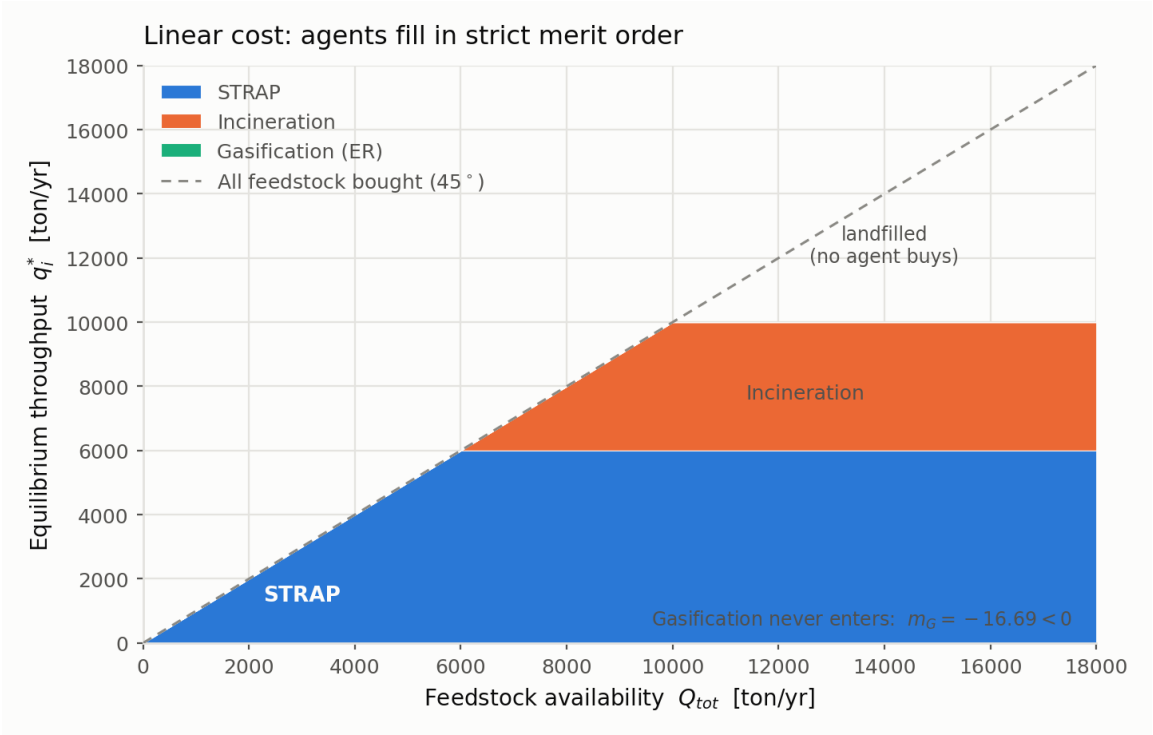



Figure 2: Equilibrium allocation against availability. STRAP fills first, then incineration; gasification never enters. Beyond 10,000 t/yr the gap to the 45° line is feedstock nobody buys.

express a scarcity *gradient* — only scarcity *regimes*. If the research question is how sensitively the competitive price responds to marginal changes in feedstock availability, this specification answers it with a flat line inside each regime, and that is a limitation of the cost structure, not of the equilibrium concept (see Section 5).

### Verification

The sweep re-solves the equilibrium at 207 points (200 evenly spaced, plus each threshold and points a hair either side of it, plus the base case). At every point:

- every KKT and complementarity condition of Eqs. (8)–(10) is satisfied with a worst residual of **exactly** 0.0 — the merit-order solution is closed-form, so there is no iteration error to accumulate;
- $\lambda$ is non-increasing in $Q_{\text{tot}}$, as an equilibrium price of a scarce good must be; and
- no two agents are ever simultaneously on the margin, so the degenerate-tie case of Section 4.1 never arises and the allocation is unique at every swept point. The sweep raises rather than reports a result if it ever does.

Produced by `Code/linear_model.py` and `Code/plots_linear.py`; full numeric output in `Code/results/linear_s`

## 5 Scope Boundaries

The following are **deliberately excluded** from this version of the model, and are not oversights:

- **Convex operating cost.** Every agent's cost here is linear ($a_i q_i$). A convex (e.g. quadratic) refinement is a possible future extension, and SA-1 sharpens the case for it: it would (i) give $\lambda$ a genuine *gradient* in $Q_{\text{tot}}$ rather than the flat treads of Figure 1, so the model could distinguish mild from severe scarcity within a regime; (ii) let an agent's *marginal* cost sit below its average cost at low throughput, which could let gasification enter the market despite $m_G < 0$ here; and (iii) remove the non-uniqueness noted in Section 4.1 whenever two agents tie on margin. Not attempted in this draft.
- **Sensitivity work beyond SA-1.** Feedstock scarcity is swept in Section 4.2. Whether the assumed plant capacities are themselves defensible — they set both thresholds in Table 4, so they matter more here than any other assumed input — and how sensitive the headline number is to the most load-bearing assumed parameters are possible follow-on work, not attempted here.
- **Output-price coupling.** All product prices $P_k$ are fixed constants throughout; no agent's throughput affects market prices.
- **Cost coupling.** $c_{\text{lf}}$ is a fixed parameter, not a function of total system-wide residue volume.
- **An emissions cap.** Emissions are computed and reported at the equilibrium but never constrain any agent's optimization.
- **Endogenous capacity / a two-stage capacity game.** The assumed capacities $\text{Cap}_i$ are fixed inputs, not the outcome of a solved simultaneous-capacity equilibrium.
- **Any second shared constraint.** Exactly one shared feedstock constraint is used in this version.

# 6 Questions for Paola

The following ten questions concern parameters used in this model that are not directly traceable to the paper, ranked roughly by how much each one moves the headline result.

1. **Incineration operating cost.** The paper's own workbook records incineration OPEX and CAPEX as literally zero, because incineration is modeled only as a third-party off-take market (the planner *sells* waste to it at 110 USD/ton), never as an owned, operated facility. This model assumed 75 USD/ton, a mid-range figure for US municipal waste-to-energy O&M. *Do you have a per-ton operating-cost figure for incineration on this feedstock, or a reference you would trust?*

2. **Incineration revenue.** Two different numbers in the authors' own notebook could both plausibly be "incineration revenue": Cwte = 259.57 USD/ton (listed alongside the STRAP resin prices as the value of the energy product) and Cgas_pw = 110 USD/ton (the price the planner is paid for *selling* waste to incineration). This model used 259.57, reasoning that an operating firm earns the product value rather than a waste-transfer fee. *Is* Cwte *the value of the energy product per tonne burned, and is* Cgas_pw *the transfer price for the waste itself?* If these are reversed, incineration's margin falls from 175.57 to 26.0 USD/ton, a factor of about 6.8.

3. **Assumed capacities.** $\text{Cap}_S = 6{,}000$, $\text{Cap}_W = \text{Cap}_G = 4{,}000$ t/yr have no basis in the paper (only location Scenario 3's mention of a 20,000 t/yr *shared* STRAP facility is suggestive). These values determine where the equilibrium price jumps as feedstock availability varies. *Is there a defensible single-plant scale for STRAP, and any basis at all for incineration or gasification capacities?*

4. **Landfill tipping fee.** This model used 60 USD/ton (a representative Wisconsin rate); the authors' own spreadsheet implies 7.836 USD/ton, roughly 8× lower than real gate rates. *Is* 7.836 *USD/ton really a tipping fee, or is it a different cost category (e.g. a per-ton* processing *cost) that was mislabeled?*

5. **Capital recovery convention.** This model used a discounted 10%/20-year CRF (0.1175); the paper's own convention is undiscounted straight-line depreciation over 20 years (CRF = 0.05). *Was the undiscounted convention a deliberate choice?* (This affects which agents are viable net of capital, but never the equilibrium throughput or $\lambda$ itself.)

6. **Feedstock price.** This model assumed feedstock is free to every agent ($p_{\text{raw}} = 0$), matching the paper's treatment of its planner. The authors' own model separately implies a transfer price of 110 USD/ton in a different context. Since $m_i$ shifts one-for-one with $p_{\text{raw}}$ for every agent (Eq. (7)), $\lambda$ would also move one-for-one with it, reaching zero around $p_{\text{raw}} = 175.57$ USD/ton. *What is the right feedstock price for a competitive multi-agent setting* — is 110 USD/ton what a recycler would actually pay, or could post-industrial waste plausibly carry a *negative* price (the generator pays to have it hauled away)?

7. **Residue/ash fractions.** The authors' spreadsheet records incineration and gasification "Waste to Disposal" as literally zero, which is not physically plausible. This model assumed $\rho_W = 0.15$ and $\rho_G = 0.08$ ton/ton from standard ranges. *Are the zero entries a placeholder, or is ash/char accounted for elsewhere in the original model?*

8. **Emissions table units.** Table 8's emissions column, headed "kg $CO_2$-eq/year," reproduces exactly if the underlying factor is actually in *tonnes* (Section 3). *Can you confirm the Table 8/9/10 emissions columns are tonnes, not kilograms?*

9. **Table 8's profit figure.** This model's reconstruction of the max-profit pathway's cost from the authors' own per-wash CAPEX/OPEX data does not reproduce the published 7,551,142 USD/yr under any single consistent accounting convention tested (shredder charged once vs. to both washes, wash-2 capital included or not). *Which capital-accounting convention produced the published Table 8 figure?*

10. **Gasification revenue inputs.** The heat price (0.13 USD/MJ) and heat yield (583 MJ/tonne) used to compute gasification's revenue both look dimensionally unusual relative to typical industrial values. *Are these in the units the notebook implies?*

## References

Muñoz-Briones, P.; Ikegwu, U. M.; Martinez, K. J.; Granger, C.; Van Lehn, R. C.; Zavala, V. M.; Huber, G. W.; Avraamidou, S. *Decision support for circular economy waste management of mixed plastic streams: A case study on post-industrial multilayer films.* (Paper on file, `Paola_Game_Theory/Paper/`.)